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CYBER SKYLINE

The National Cyber League
A Community Where Cybersecurity Is a Passion

Jonathan Hemingway

NCL Spring 2025 Team Game Scouting Report

Dear Jonathan Hemingway (Team "SANS.edu Sentinels 391"),

Thank you for participating in the National Cyber League (NCL) Spring 2025 Season! Our goal is to prepare the next generation of cybersecurity professionals, and your participation is helping achieve that goal.

The NCL was founded in May 2011 to provide an ongoing virtual training ground for collegiate students to develop, practice, and validate their cybersecurity skills in preparation for further learning, industry certifications, and career readiness. The NCL scenario-based challenges were designed around performance-based exam objectives of CompTIA certifications and are aligned to the National Initiative for Cybersecurity Education (NICE) Cybersecurity Workforce Framework published by the National Institute of Standards and Technology (NIST).

As you look to a future career in cybersecurity, we hope you find this report to be valuable in both validating skills and identifying areas for improvement across the nine NCL skills categories. You can use this NCL Scouting Report to:

- Validate your skills to employers in any job application or professional portfolio;
- Show case your achievements and strengths by including the Score Card view of your performance as part of your résumé or simply sharing the validation link so that others may view the detailed version of this report.

The NCL Spring 2025 Season had 9,216 students/players and 596 faculty/coaches from 510 two- and four-year schools & 288 high schools across all 50 U.S. states registered to play. The Individual Game Capture the Flag (CTF) event took place from April 11 through April 13. The Team Game CTF event took place from April 25 through April 27. The games were conducted in real-time for students across the country. You were in the Experienced Students Bracket, consisting of students enrolled in advanced degrees or hold extensive industry working experience.

NCL is powered by Cyber Skyline's cloud-based skills evaluation platform. Cyber Skyline hosted the scenario-driven cybersecurity challenges for players to compete and track their progress in real-time.



To validate this report, please access: cyberskyline.com/report/4E6KW7AT58MJ

Congratulations for your participation in the NCL Spring 2025 Team Game! We hope you will continue to develop your knowledge and skills and make meaningful contributions as part of the Information Security workforce!

Dr. David Zeichick
NCL Commissioner



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CYBER SKYLINE

**EXPERIENCED
STUDENTS RANK
4TH PLACE
OUT OF 457
PERCENTILE
100TH**

NATIONAL CYBER LEAGUE SCORE CARD

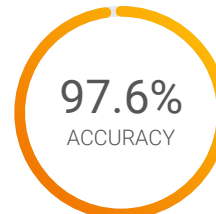
NCL SPRING 2025 TEAM GAME

YOUR TOP CATEGORIES

**ENUMERATION &
EXPLOITATION
99TH PERCENTILE**

**WEB APPLICATION
EXPLOITATION
99TH PERCENTILE**

**CRYPTOGRAPHY
99TH PERCENTILE**



Average: 66.4%

cyberskyline.com/report/4E6KW7AT58MJ



NCL Spring 2025 Team Game

The NCL Team Game is designed for student players nationwide to compete in realtime in the categories listed below. The Team Game promotes camaraderie and evaluates the collective technical cybersecurity skills of the team members.

4TH PLACE OUT OF 457 EXPERIENCED STUDENTS RANK	3000 POINTS OUT OF 3000 PERFORMANCE SCORE	97.6% ACCURACY	100.0% COMPLETION
100 th Experienced Students Percentile	Average: 1594.9 Points	Average: 66.4%	Average: 62.4%
Cryptography Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.	325 POINTS OUT OF 325	89.5% ACCURACY	COMPLETION: 100.0%
Enumeration & Exploitation Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.	380 POINTS OUT OF 380	100.0% ACCURACY	COMPLETION: 100.0%
Forensics Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.	300 POINTS OUT OF 300	100.0% ACCURACY	COMPLETION: 100.0%
Log Analysis Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.	315 POINTS OUT OF 315	100.0% ACCURACY	COMPLETION: 100.0%
Network Traffic Analysis Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.	300 POINTS OUT OF 300	100.0% ACCURACY	COMPLETION: 100.0%
Open Source Intelligence Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.	350 POINTS OUT OF 350	96.4% ACCURACY	COMPLETION: 100.0%
Password Cracking Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.	320 POINTS OUT OF 320	100.0% ACCURACY	COMPLETION: 100.0%
Scanning & Reconnaissance Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.	300 POINTS OUT OF 300	100.0% ACCURACY	COMPLETION: 100.0%
Web Application Exploitation Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.	310 POINTS OUT OF 310	100.0% ACCURACY	COMPLETION: 100.0%

Note: Survey module (100 points) was excluded from this report.



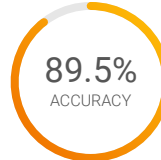


Cryptography Module

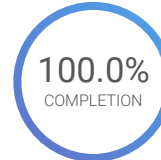
Identify techniques used to encrypt or obfuscate messages and leverage tools to extract the plaintext.

6TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

325 POINTS
OUT OF 325
PERFORMANCE SCORE



Average: 67.3%



Average: 59.0%

TOP NICE WORKROLES

Security Control Assessor
Secure Software Assessor
Exploitation Analyst
Cyber Operator
Security Architect

99th Experienced Students
Percentile

Average: 155.0 Points

So Basic (Easy)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: 100.0%

Decode the plaintext of messages encoded with common number bases

For Fun (Easy)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: 100.0%

Decrypt messages that are encrypted using different shift ciphers

Crypto Therapy (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: 100.0%

Exploit a poor implementation of RSA to decrypt sensitive data

Skip To My Lou (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: 100.0%

Decrypt a message that was encrypted using the Skipjack cipher

Infinite Cosmic Power (Hard)

125 POINTS
OUT OF 125

77.8%
ACCURACY

COMPLETION: 100.0%

Perform a differential power attack on a power trace to obtain an AES encryption key





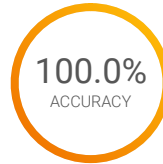
Enumeration & Exploitation Module

Identify actionable exploits and vulnerabilities and use them to bypass the security measures in code and compiled binaries.

5TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

380 POINTS
OUT OF 380
PERFORMANCE SCORE

Average: 179.0 Points



Average: 71.9%



Average: 65.2%

TOP NICE WORKROLES

Cyber Operator
Target Developer
Exploitation Analyst
Software Developer
Systems Security Analyst

99th Experienced Students
Percentile

Ransomed (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze source code to bypass an XOR-based authentication mechanism

Crawling (Medium)

150 POINTS
OUT OF 150

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform a privilege escalation attack by coercing a binary with elevated permissions into running untrusted software

ExploitExhibit (Hard)

130 POINTS
OUT OF 130

100.0%
ACCURACY

COMPLETION: **100.0%**

Exploit a C++ program by undermining the heap memory or leaking vtable pointers

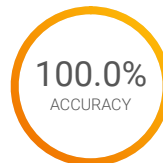
Forensics Module

Utilize the proper tools and techniques to analyze, process, recover, and/or investigate digital evidence in a computer-related incident.

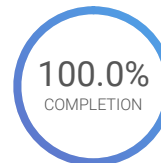
7TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

300 POINTS
OUT OF 300
PERFORMANCE SCORE

Average: 188.5 Points



Average: 69.7%



Average: 66.3%

TOP NICE WORKROLES

Cyber Defense Forensics
Analyst
Cyber Crime Investigator
Cyber Defense Incident
Responder
Cyber Defense Analyst

Origin Story (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Investigate a Git repository to find leaked secrets

Ransom Residue (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform a forensic investigation of a computer that was compromised by ransomware by analyzing a disk image file

Archive (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Automate the extraction of files by matching file signatures in a raw byte stream





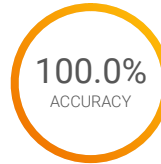
Log Analysis Module

Utilize the proper tools and techniques to establish a baseline for normal operation and identify malicious activities using log files from various services.

7TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

315 POINTS
OUT OF 315
PERFORMANCE SCORE

Average: 206.5 Points



Average: 62.0%



Average: 69.2%

TOP NICE WORKROLES

Cyber Defense Analyst
Systems Security Analyst
All-Source Analyst
Cyber Defense Forensics Analyst
Data Analyst

What Remains (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze event logs from a Windows 7 workstation that was compromised by EternalBlue

Compromising (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a NetFlow log to identify compromised devices and malicious behavior

Coords (Hard)

115 POINTS
OUT OF 115

100.0%
ACCURACY

COMPLETION: **100.0%**

Parse a custom binary file format and perform geospatial analysis of drone flight paths

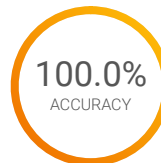
Network Traffic Analysis Module

Identify malicious and benign network traffic to demonstrate an understanding of potential security breaches.

21ST PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

300 POINTS
OUT OF 300
PERFORMANCE SCORE

Average: 197.6 Points



Average: 75.2%



Average: 78.7%

TOP NICE WORKROLES

Cyber Defense Analyst
All-Source Analyst
Cyber Defense Incident Responder
Target Network Analyst
Cyber Operator

HTTP Forever (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a packet capture with HTTP traffic to identify web browsing behavior

SQL Storage Secrets (Medium)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze PostgreSQL traffic to discover information about a database and the binary content inside it

Ping of Domin8tion (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Analyze a packet capture to extract an image that was exfiltrated by encoding data in ICMP packet headers



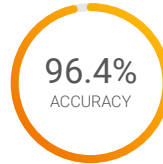


Open Source Intelligence Module

Utilize publicly available information such as search engines, public repositories, social media, and more to gain in-depth knowledge on a topic or target.

7TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

350 POINTS
OUT OF 350
PERFORMANCE SCORE



Average: 63.1%



Average: 76.5%

TOP NICE WORKROLES
Systems Security Analyst
Target Developer
System Administrator
Research & Development
Specialist
Cyber Intel Planner

99th Experienced Students
Percentile

Average: 233.4 Points

Rules of Conduct (Easy)

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Introductory challenge on acceptable conduct during NCL

Swap (Easy)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Utilize blockchain tools to perform an analysis of cryptocurrency transactions

Granny (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: **100.0%**

Use open source tools to find information about the subject of an image

Sla.cking (Medium)

60 POINTS
OUT OF 60

100.0%
ACCURACY

COMPLETION: **100.0%**

Perform introspection on deep web forums using the Internet Archive

Gamma Brasiliensis (Hard)

100 POINTS
OUT OF 100

83.3%
ACCURACY

COMPLETION: **100.0%**

Conduct an open source investigation on an individual to create a timeline of their actions and cryptocurrency transactions



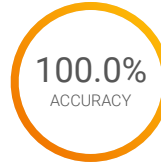


Password Cracking Module

Identify types of password hashes and apply various techniques to efficiently determine plaintext passwords.

24TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

320 POINTS
OUT OF 320
PERFORMANCE SCORE



Average: 88.6%



Average: 74.5%

TOP NICE WORKROLES

Cyber Operator
Exploitation Analyst
Systems Security Analyst
Cyber Defense Incident
Responder
Cyber Crime Investigator

95th Experienced Students
Percentile

Average: 226.0 Points

Loaded Hash Browns (Easy)

30 POINTS
OUT OF 30

100.0%
ACCURACY

COMPLETION: **100.0%**

Generate password hashes using MD5, SHA1, and SHA256

You're So Basic (Easy)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Crack MD5 password hashes for commonly used passwords

786b63642e636f6d2f393336 (Medium)

45 POINTS
OUT OF 45

100.0%
ACCURACY

COMPLETION: **100.0%**

Crack Windows NTLM password hashes that may not be found in common rainbow tables

Zip (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: **100.0%**

Crack the insecure password for a protected zip file

Crunch Time (Medium)

50 POINTS
OUT OF 50

100.0%
ACCURACY

COMPLETION: **100.0%**

Create a custom wordlist to crack passwords based on known password rules

Router (Hard)

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Build a custom wordlist to crack passwords by augmenting permutation rules using known password complexity requirements





Scanning & Reconnaissance Module

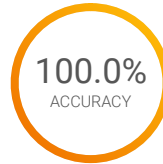
Identify and use the proper tools to gain intelligence about a target including its services and potential vulnerabilities.

7TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

99TH Experienced Students
Percentile

300 POINTS
OUT OF 300
PERFORMANCE SCORE

Average: 143.7 Points



Average: 56.0%



Average: 48.1%

TOP NICE WORKROLES

Vulnerability Assessment
Analyst
Target Network Analyst
Cyber Operations Planner
Target Developer
Security Control Assessor

LuckyCat (Easy)

Classify malware from benign samples by creating custom Yara rules

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Schrödinger's Port (Medium)

Disable a local firewall and perform a port scan on an internal network

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Ephemeral (Hard)

Scan a device and perform a port knocking sequence to access a hidden service

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Web Application Exploitation Module

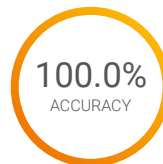
Identify actionable exploits and vulnerabilities and use them to bypass the security measures in online services.

5TH PLACE
OUT OF 457
EXPERIENCED STUDENTS RANK

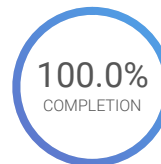
99TH Experienced Students
Percentile

310 POINTS
OUT OF 310
PERFORMANCE SCORE

Average: 136.2 Points



Average: 84.7%



Average: 46.4%

TOP NICE WORKROLES

Cyber Operator
Software Developer
Exploitation Analyst
Systems Security Analyst
Database Administrator

Doohickey Divulgence (Easy)

Perform analysis on minified source code to decode and extract sensitive information

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Pierre's Store (Medium)

Perform a SQL injection attack to leak sensitive information from a web application

100 POINTS
OUT OF 100

100.0%
ACCURACY

COMPLETION: **100.0%**

Cash Rules Everything Around Me (Hard)

Perform a series of cache deception attacks against a web application

110 POINTS
OUT OF 110

100.0%
ACCURACY

COMPLETION: **100.0%**

